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CA2: Final Interim Report

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1. Background

Singapore Polytechnic is a post-secondary education institution in Singapore that emphasizes hands-on, industry-relevant education. As part of its continuous efforts in digital transformation, the institution has explored the use of emerging technologies to enhance teaching and learning processes. One key area of focus has been the applied use of artificial intelligence, particularly Large Language Models (LLMs), to support educators in educational content creation.

During my year-long internship at Singapore Polytechnic, I was involved in a project named AgentSoC, which focuses on leveraging agentic AI systems to generate educational materials. AgentSoC is an AI-driven software development project designed to support lecturers and teaching staff by automating the generation of lecture slides and laboratory exercises from provided source materials such as books, documents, and online resources. The project aims to reduce the workload associated with content preparation while maintaining educational relevance and quality. My role in the project involved supporting the exploration and implementation of AI-based approaches for educational material generation, as well as working with the system architecture and deployment considerations of an agentic AI system.

2. Solution Development

2.1 Overview of the AgentSoC System

AgentSoC is an agentic AI system designed to support lecturers and teaching staff by automating the generation of educational materials such as lecture slides and laboratory exercises. The system operates by processing source materials provided by educators and leveraging LLMs to generate structured educational content. As the project involves an agentic AI architecture, the reliability and behaviour of the underlying language model play a critical role in overall system performance.

My involvement in the solution development phase focused on experimentation with language models, understanding how these models integrate with an agentic system, and supporting deployment activities. While a separate student project group was responsible for broader system development, my work was conducted independently under the guidance of supervisors, with a focus on technical exploration and system integration rather than full application ownership.

2.2 Language Model Experimentation

A key component of the solution development process involved experimenting with large language models to assess their suitability for use within an agentic AI system. Since agentic systems rely heavily on the stability, consistency, and controllability of model outputs, experimentation was necessary to evaluate whether open-source models could serve as viable alternatives to paid proprietary models.

During this phase, I explored the use of open-source language models such as Phi-4 and Llama 3.1 and investigated the feasibility of fine-tuning these models using the Unsloth library. The motivation for this approach was to reduce dependency on paid APIs while maintaining acceptable output quality for educational content generation. Experiments were conducted using cloud-based GPU resources provided by Hugging Face Spaces, allowing iterative testing and refinement of model behaviour.

However, fine-tuning large language models proved challenging due to the limited availability of high-quality training data. As a result, although experimentation provided valuable insights into model behaviour and limitations, the project ultimately adopted an alternative language model solution, Gemini, which was more stable and better supported for the project's requirements.

2.3 Experimentation Approach and Techniques

Given the experimental nature of language model fine-tuning, a trial-and-error methodology was adopted throughout the development process. Model configurations were adjusted iteratively in response to errors, unexpected outputs, or performance limitations. Discussions with supervisors formed a central part of the problem-solving process, as formal documentation and online resources were often insufficient due to the emerging state of the field.

Additional techniques included reviewing error outputs, modifying parameters, adjusting configurations, and re-running experiments to observe changes in behaviour. This iterative approach allowed progressive refinement of experimental setups and supported informed decision-making despite technical uncertainty.

2.4 Testing Approach

Testing during the solution development phase was exploratory in nature. Rather than implementing a formal testing framework, model outputs and system behaviour were manually observed to assess feasibility and consistency. This approach was appropriate given the exploratory focus of the work and the absence of stable, production-ready models at this stage. Formal testing mechanisms were identified as a potential area for future improvement but were outside the scope of the internship.

2.5 Deployment and Integration

Deployment of the AgentSoC system was carried out using the Render cloud platform. I was responsible for configuring environment variables, setting up the required services, and resolving build or startup issues during deployment. The system was successfully deployed and verified to be running on the platform.

Following deployment, issues were encountered due to outdated external language model API links, which affected full system functionality. Although efforts were made to address these issues, complete resolution was not achieved due to external dependency constraints and project scope limitations. Nevertheless, the deployment process provided valuable experience in real-world system integration and highlighted the challenges of maintaining AI systems that rely on rapidly changing external services.

3. Evaluate Findings

3.1 Things That Went Well

Several aspects of the project progressed positively during the internship. The deployment of the AgentSoC system to the Render cloud platform was successful, and the application was able to run as intended. In addition, the deployment process was documented, which helped capture practical knowledge for future reference and reduced the need for subsequent contributors to repeat the same setup steps.

From a learning perspective, the internship provided valuable exposure to deploying cloud-based environments for large language model experimentation, particularly when using GPU resources for fine-tuning exploration with Unsloth. I also gained a deeper understanding of agentic AI systems, large language models, and how they are integrated into real-world applications. Furthermore, my familiarity with version control practices improved through consistent use of Git across GitHub and GitLab during development and deployment.

3.2 Challenges Encountered

Despite these successes, several challenges were encountered. Unsloth is a relatively new tool, and the lack of comprehensive documentation and community support made debugging difficult. As a result, resolving errors often required additional experimentation and discussion rather than relying on existing references.

Fine-tuning large language models using Unsloth also proved to be more challenging than initially expected, as the resulting model behaviour was often inconsistent and did not meet the desired level of stability. Additionally, the overall system lacked a formal testing layer, as the student project group focused primarily on implementation. Due to time constraints, the approaching end of the internship, and limited exposure to large-scale testing frameworks, it was not feasible to design and implement a comprehensive testing strategy within the internship period.

3.3 Problems Encountered and Decisions Made

In response to these challenges, several key decisions were made. The necessity of fine-tuning language models using Unsloth was re-evaluated, given that agentic AI systems can achieve reasonable robustness when paired with more stable and well-supported language models. As a result, further fine-tuning efforts were deprioritised.

Due to limited time, focus was shifted toward documenting findings, challenges, and deployment processes for supervisors and future interns. This ensured continuity and reduced the likelihood that future contributors would need to start from scratch. Scope was also narrowed to prioritise deployment, system understanding, and knowledge transfer over further experimentation.

3.4 Changes to Original Scope or Plans

Initially, the project scope included exploring fine-tuning techniques to identify or improve the most suitable language model for AgentSoC. However, this scope shifted as practical constraints emerged. The focus moved away from deep fine-tuning and toward maintaining and supporting the AgentSoC system, including deployment and debugging efforts. This change allowed the project to deliver practical value despite unresolved technical limitations.

3.5 Potential Improvements

Several potential improvements were identified during the internship. Documentation could be further expanded to include more detailed troubleshooting guidance and architectural explanations. Developing stronger debugging strategies and technical knowledge would also improve efficiency when working with emerging AI tools. In addition, more consistent and structured version control practices using Git could further enhance collaboration and maintainability.

3.6 Next Steps and Follow-Up Actions

A key next step for AgentSoC is the implementation of a formal testing layer. Comprehensive testing is critical to ensuring system reliability, especially for an agentic AI system that relies on external language models and evolving APIs. Establishing automated or structured testing processes would significantly improve system stability and support future development efforts.

4. Conclusion

Overall, the internship provided valuable experience in applying AI technologies within an educational context through the AgentSoC project. The project explored how agentic AI systems and large language models can support lecturers and teaching staff by reducing the workload associated with educational content creation, while also highlighting practical challenges involved in deploying AI-driven systems.

Although not all technical objectives were fully realised, the internship successfully met its learning and exploratory goals. I gained hands-on experience with agentic AI architectures, language model experimentation, and cloud deployment, as well as a deeper understanding of system stability, documentation, and real-world constraints. These experiences provided a strong foundation for future work in AI system development and deployment.

References

About Singapore Polytechnic:

https://en.wikipedia.org/wiki/Singapore_Polytechnic

Unsloth:

<https://github.com/unslothai/unsloth>

Render:

<https://render.com/>

Hugging Face:

<https://huggingface.co/spaces>

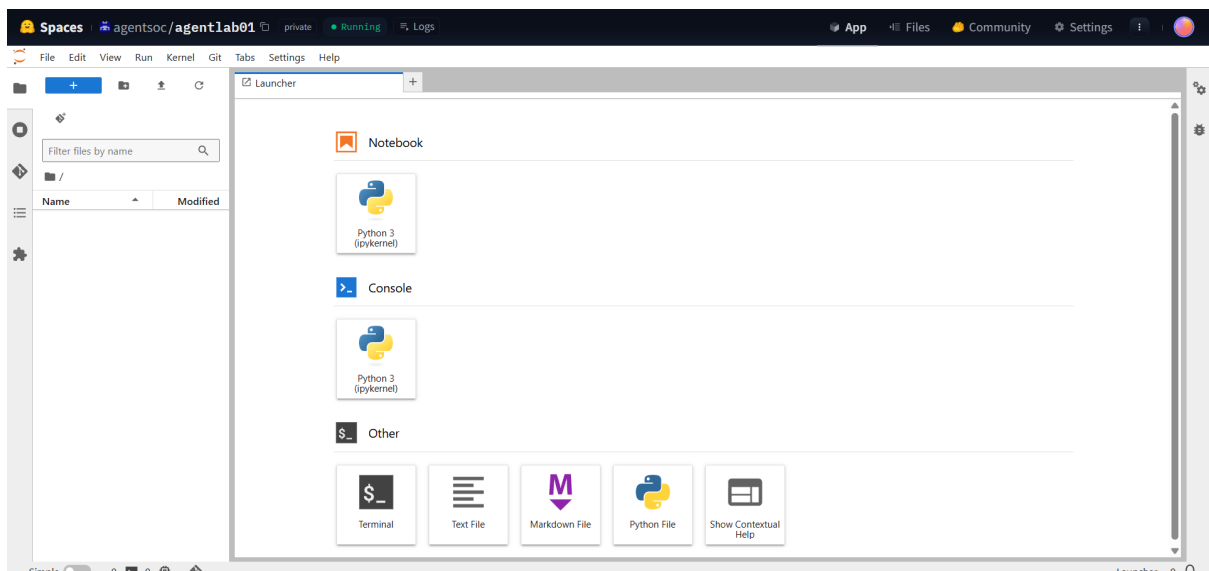
Soft Prompt Files:



phi_4.ipynb

Appendix

Appendix A: Hugging Face Space JupyterLab



Appendix B: Hugging Face Spaces JupyterLab Repository (GitHub)

The screenshot shows a GitHub repository for user 'dacihi' with the name 'testing'. The repository has 6 commits, last updated 2 weeks ago (commit b30c557). The repository is empty, with no description, website, or topics provided. The right sidebar shows repository statistics: 0 stars, 0 watching, and 1 fork. The 'Releases' section indicates no releases are published, and the 'Packages' section indicates no packages are published. The 'Languages' section shows a progress bar for 'Jupyter Notebook' at 100.0%.

File	Commit Message	Commit Date
content	Initial commit	4 months ago
data	Initial commit	4 months ago
ppt_generated	Initial commit	4 months ago
.gitattributes	Initial commit	4 months ago
.gitignore	add .ssh to .gitignore and deleted the git_test.txt	2 weeks ago
LLama_3.1_8B.ipynb	Initial commit	4 months ago
Project.jupyterlab-workspace	Initial commit	4 months ago
Versions Check.ipynb	Initial commit	4 months ago
default.jupyterlab-workspace	Initial commit	4 months ago
gittest.txt	testing	2 weeks ago
ollama.ipynb	Initial commit	4 months ago
phi_4.ipynb	Initial commit	4 months ago
slides_alpacajsonl	Initial commit	4 months ago
softphi4-Copy1.ipynb	Initial commit	4 months ago
softphi4.ipynb	Initial commit	4 months ago

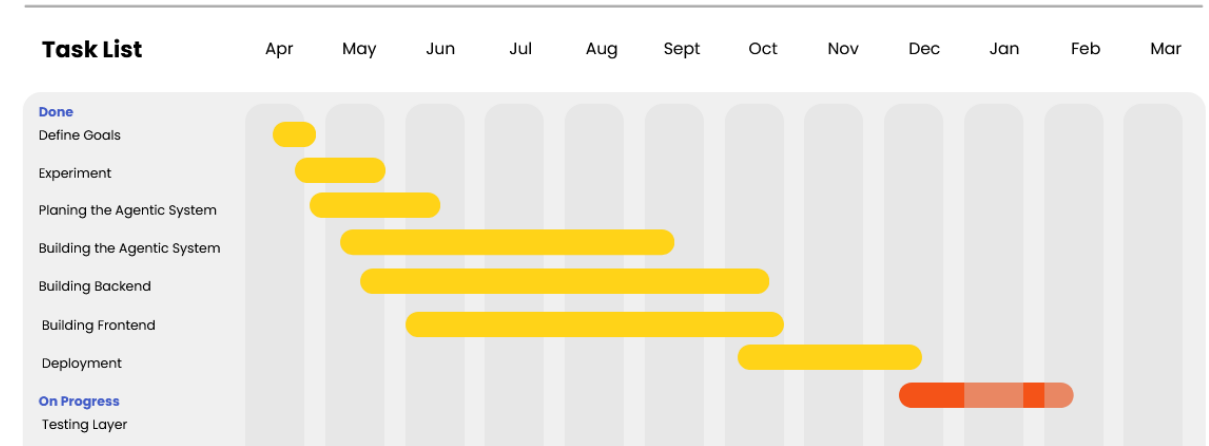
Appendix C: Render Dashboard

The screenshot shows the Render dashboard for the 'AgentSOC' project. The dashboard is divided into sections: 'Dashboard', 'AgentSOC', 'Overview', 'MANAGE', and 'Settings'. The 'Overview' section is currently selected. The 'Production' section shows a table of services with columns: SERVICE NAME, STATUS, RUNTIME, REGION, and UPDATED. The table lists four services: redis.6, agentsoc, agentsoc-agentic-backend, and agentsoc-agentic-agentv2. All services are 'Suspended by you' and are running on 'Docker' in the 'Singapore' region. The 'UPDATED' column shows the last update time for each service: 1mo, 2mo, 3mo, and 3mo respectively. There is a search bar for resources in Production and a '+ Add environment' button.

SERVICE NAME	STATUS	RUNTIME	REGION	UPDATED
redis.6	Suspended by you	Valkyrie 8	Singapore	1mo
agentsoc	Suspended by you	Docker	Singapore	2mo
agentsoc-agentic-backend	Suspended by you	Docker	Singapore	3mo
agentsoc-agentic-agentv2	Suspended by you	Docker	Singapore	3mo

Appendix D: Project Timeline

Project Timeline AgentSoC



Appendix E: AgentSoC Website Ai Chatting

